ST102 Statistics — Problem Set 6: Hypothesis Testing

1. A machine fills bags of flour. The fill weight is normal with known standard deviation $\sigma = 12\,\text{g}$. A random sample of $n = 36$ bags has mean $\bar{x} = 496\,\text{g}$. Test $H_0 : \mu = 500$ against $H_1 : \mu \neq 500$ at the 5% level.

2. The times (in minutes) that eight students needed for a quiz were

   $$12,\ 15,\ 11,\ 14,\ 13,\ 16,\ 12,\ 15.$$

   Assuming normality, test at the 5% level whether the mean time exceeds 12 minutes.

3. A coin is tossed 100 times and shows 60 heads. Is the coin fair? Use $\alpha = 0.05$.

4. True or false? Explain briefly.

   (a) The p-value is the probability that H_0 is true.

   (b) If H_0 is rejected at the 1% level, it is also rejected at the 5% level.

   (c) For a fixed true effect and fixed α, increasing n increases the power of the test.

   (d) If a 95% confidence interval for μ does not contain μ_0, the two-sided test of $H_0 : \mu = \mu_0$ rejects at the 5% level.

5. $X \sim N(\mu, 15^2)$, $n = 25$. We test $H_0 : \mu = 100$ against $H_1 : \mu > 100$ at $\alpha = 0.05$. Find the rejection region for $\bar{X}$ and the power of the test when $\mu = 106$. What is the probability of a Type II error?

Answers (for self-check). 1. $z = -0.33$, do not reject H_0. 2. $t \approx 2.39$, reject H_0. 3. see class. 4. F, T, T, T. 5. Reject if $\bar{X} > 104.9$; power ≈ 0.64; $\beta \approx 0.36$.